# Von Mises Transformation

11/5/25

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## Results and Discussion

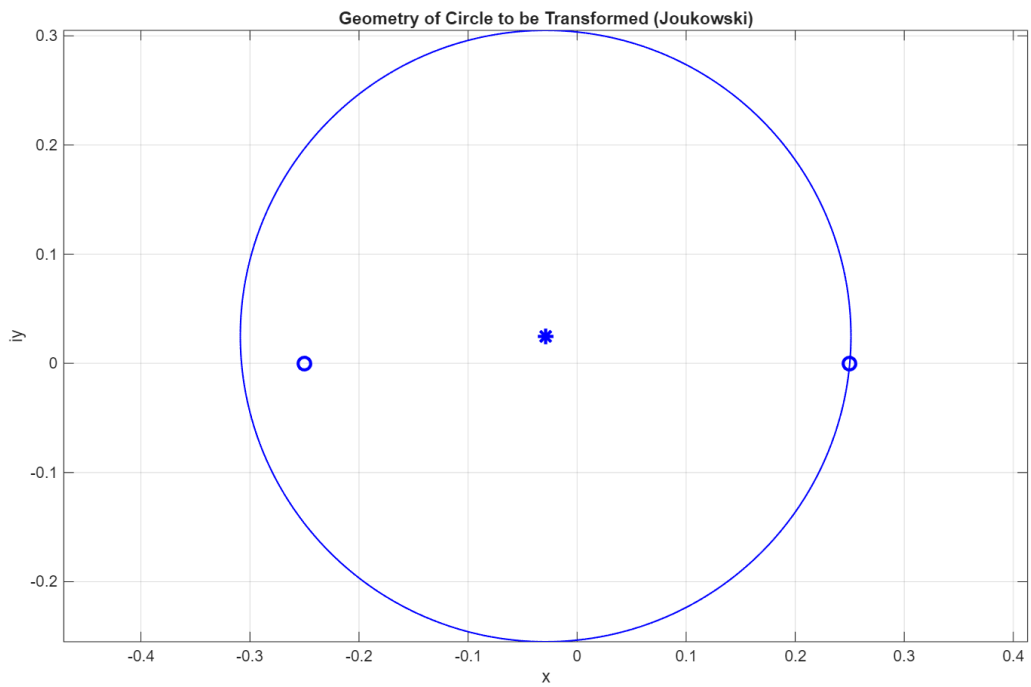


*Figure 1. Circle Geometry for Joukowski Airfoil.*

Figure 1 demonstrates the geometry of the circle needed to transform the Joukowski airfoil, which has zeros at -0.25 and 0.25 and center location at -0.0289 + 0.0250i. Equation 1 is imperative in calculating the transformation circle:

$$\zeta = a\mathrm{e}^{\mathrm{i}\theta} + \mu \tag{1}$$

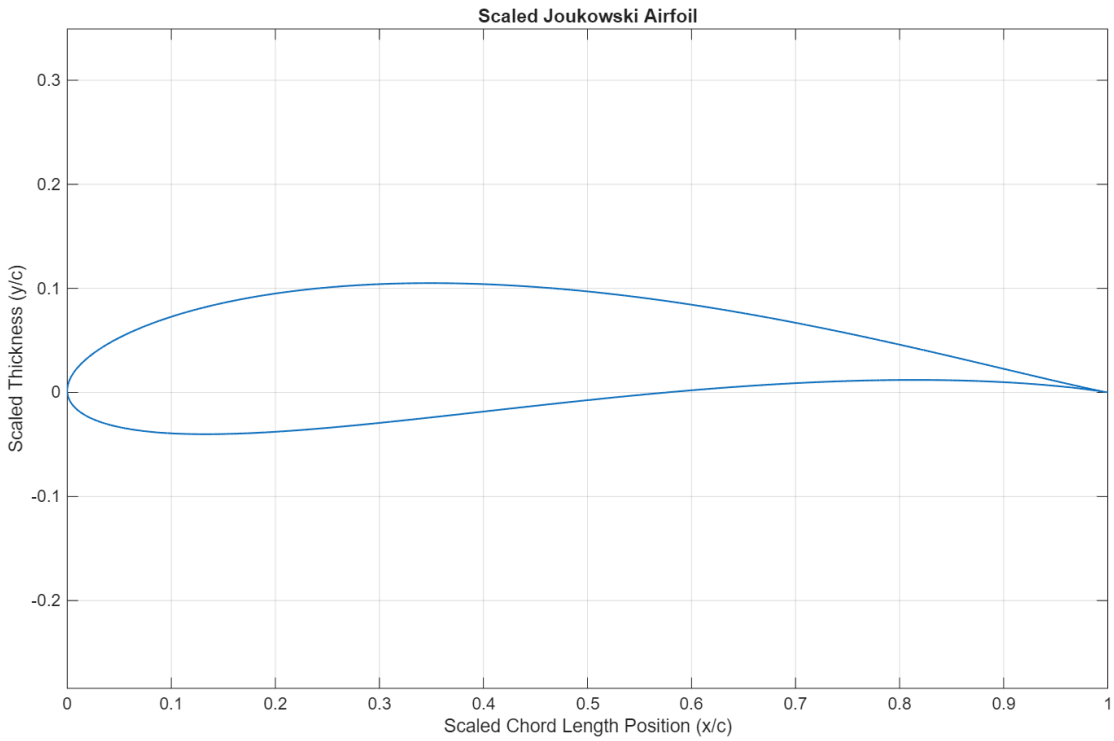


*Figure 2. Joukowski Airfoil Transformation.*

Figure 2 shows the Joukowski airfoil transformation from the circle in Figure 2. This airfoil has a thickness ratio of roughly 15% and camber ratio of roughly 5%. Equation 2 was key in the transformation of the circle geometry to the Joukowski airfoil pictured above:

$$z = \zeta + \frac{\mathrm{b}^2}{\zeta} \tag{2}$$

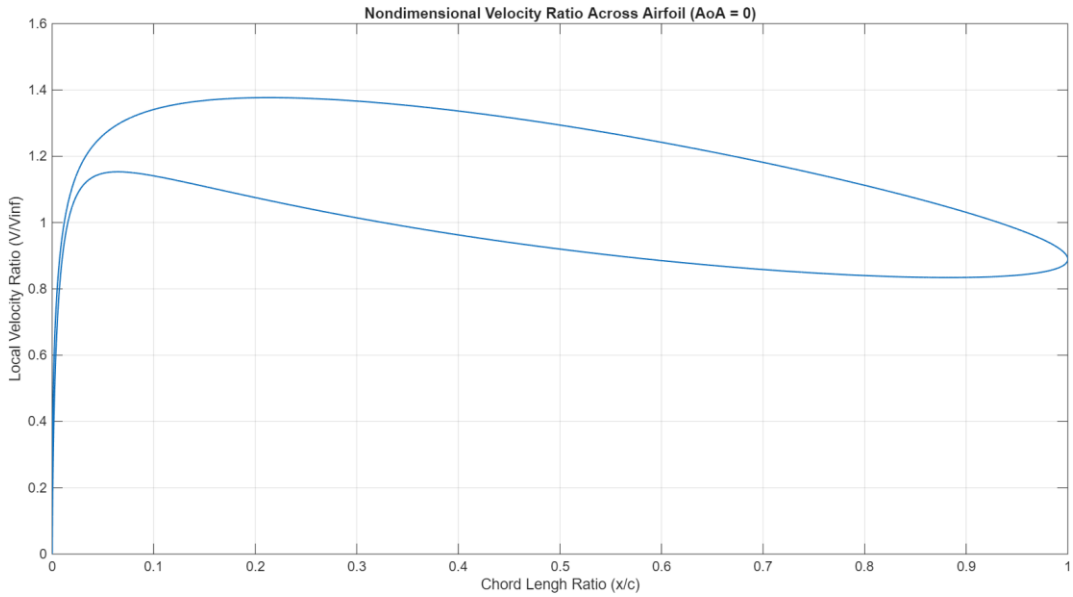


*Figure 3. Velocity Ratio at Angle of Attack of 0 Degrees*

Figure 3 illustrates the local velocity ratio for the Joukowski airfoil in Figure 2 at an angle of attack of zero degrees. Equation 3 coupled with the kutta condition is crucial in calculating the velocity distribution:

$$\frac{V}{V_\infty} = \frac{2[sin(\alpha+\beta) - sin(\alpha-\theta)]}{\left|\frac{dz}{d\zeta}\right|_{\text{on circle}}} \tag{3}$$

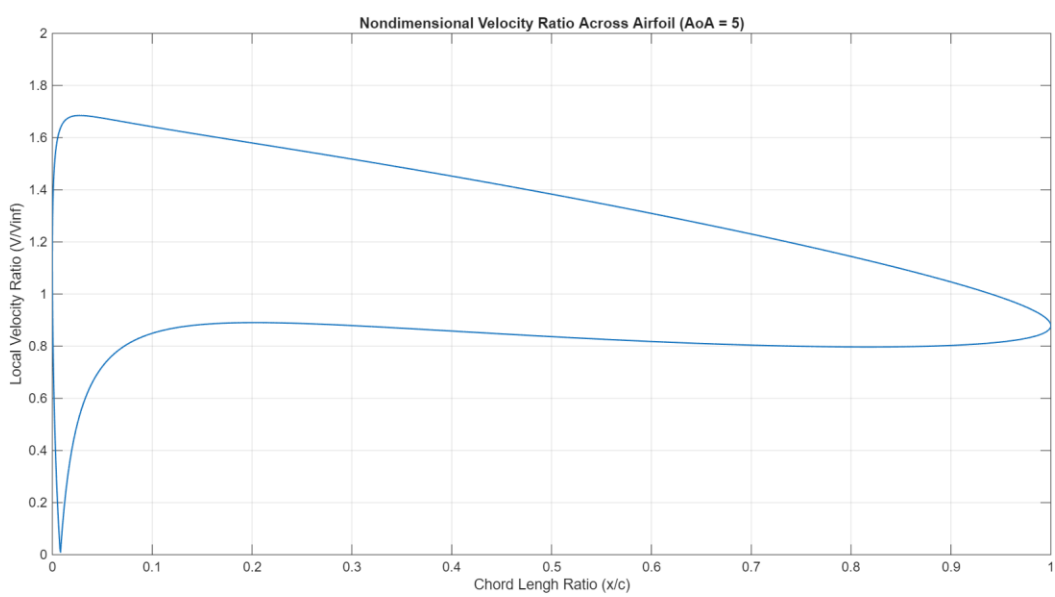


*Figure 4. Velocity Ratio at Angle of Attack of 5 Degrees.*

Figure 4 demonstrates the velocity ratio of the Joukowski airfoil at an angle of attack of 5 degrees, utilizing Equation 3 as well.

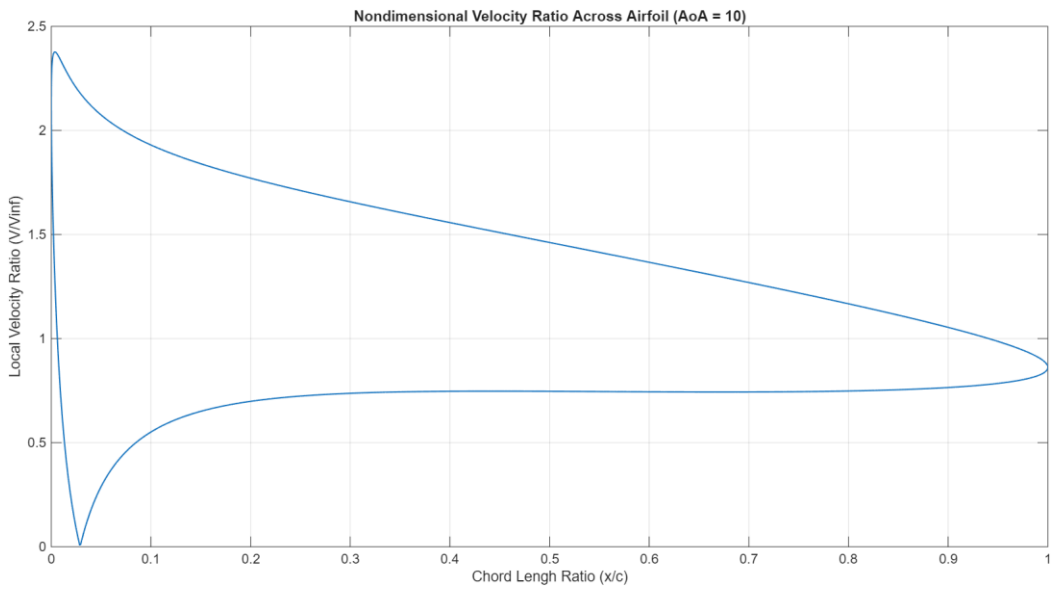


*Figure 5. Velocity Ratio at Angle of Attack of 10 Degrees.*

Figure 5 shows the velocity ratio of the Joukowski airfoil at an angle of attack of 10 degrees, utilizing Equation 3, once again.

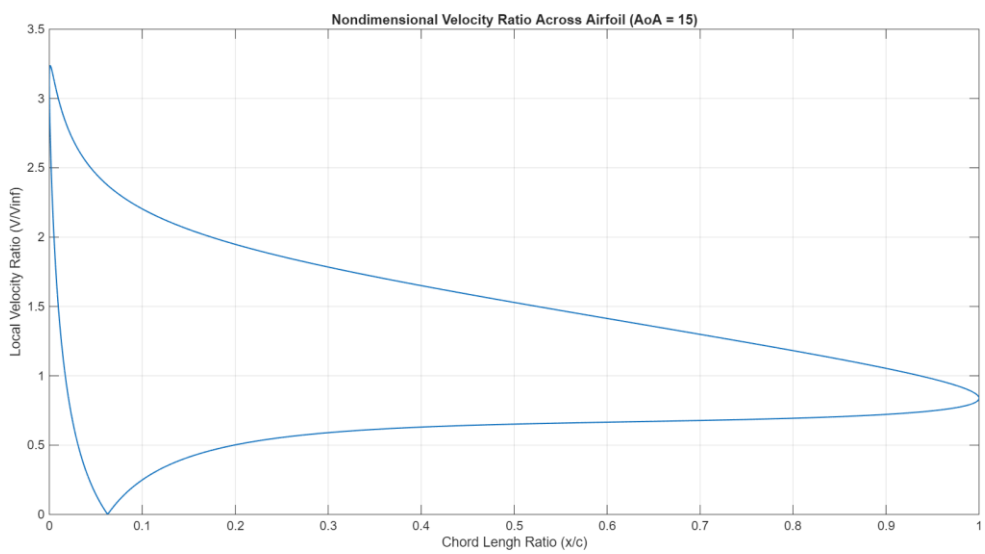


*Figure 6. Velocity Ratio at Angle of Attack of 15 degrees.*

Figure 6 illustrates the velocity ratio across the Joukowski airfoil at an angle of attack of 15 degrees, using Equation 3 as well.

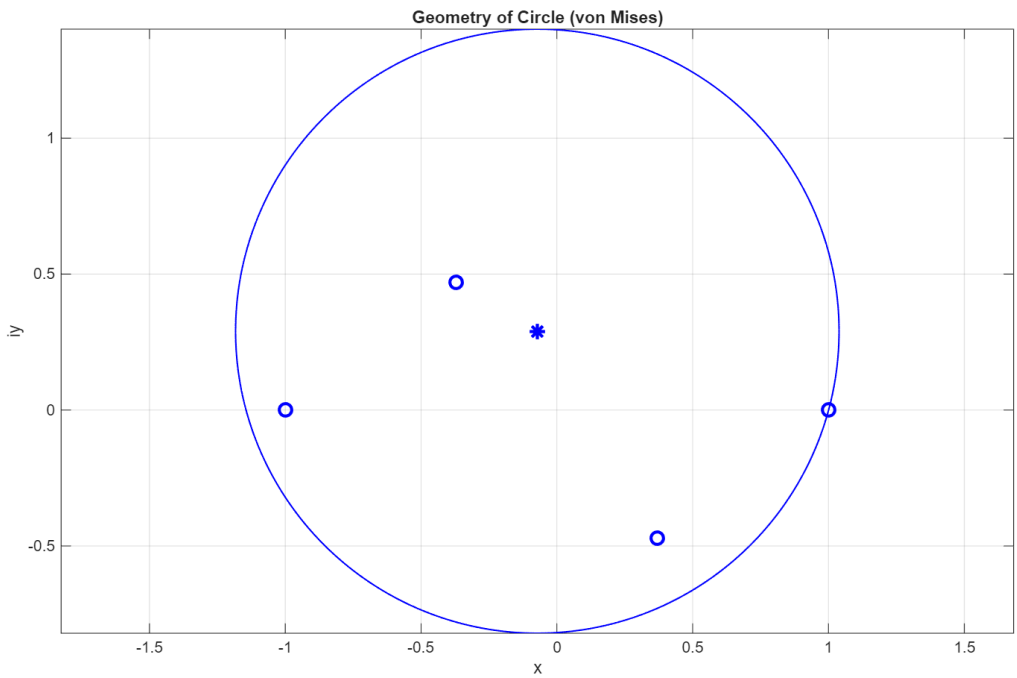


*Figure 7. Circle Geometry for von Mises Airfoil.*

Figure 7 demonstrates the geometry of the circle needed to transform the von Mises Airfoil. This circle geometry represents the first von Mises example given in the chapter 4 notes. As such, it has zeros at ZT = 1.0, Z2 = -1.0, Z3 = 0.370 – 0.47i, and Z4 = -0.37 + 0.47i. The center circle is located at -0.072 + i0.290. The circle geometry still utilizes Equation 1.

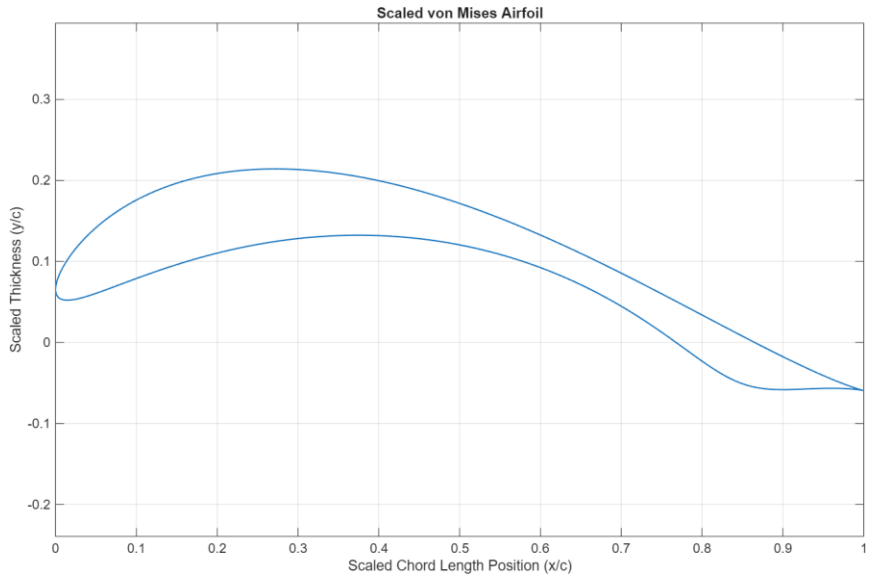


*Figure 8. von Mises Airfoil Transformation.*

Figure 8 represents the transformation for the von Mises Airfoil. This transformation relies on Equation 4:

$$z = \zeta + \frac{c_1}{\zeta^1} + \frac{c_2}{\zeta^2} + \frac{c_3}{\zeta^3} \tag{4}$$

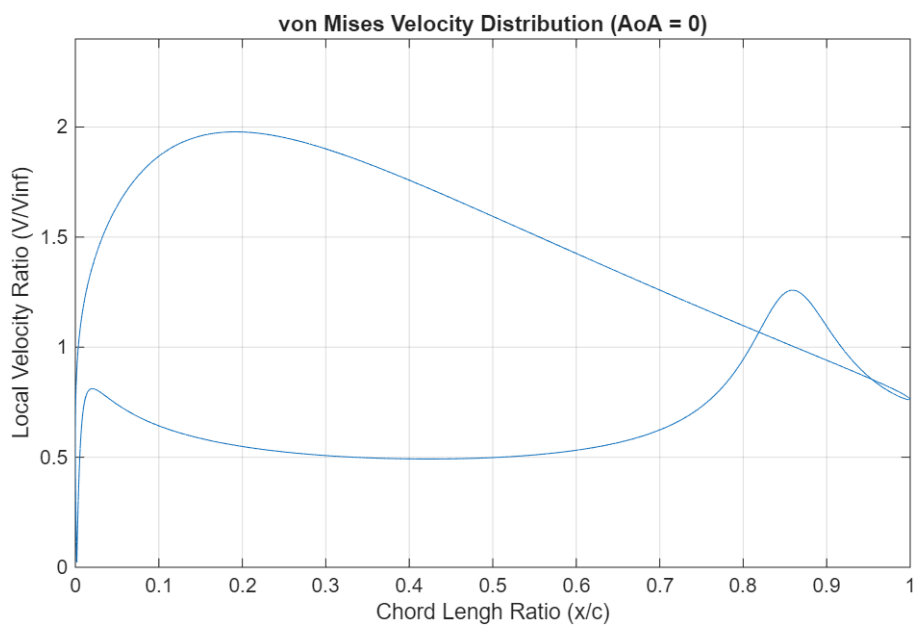


*Figure 9. Velocity Ratio at Angle of Attack of 0 Degrees.*

Figure 9 illustrates the velocity ratio across the von Mises airfoil at an angle of attack of zero degrees, using the kutta condition at the trailing edge to get smooth flow. The velocity distribution is calculated using Equation 3, making sure to use z for the von Mises case.

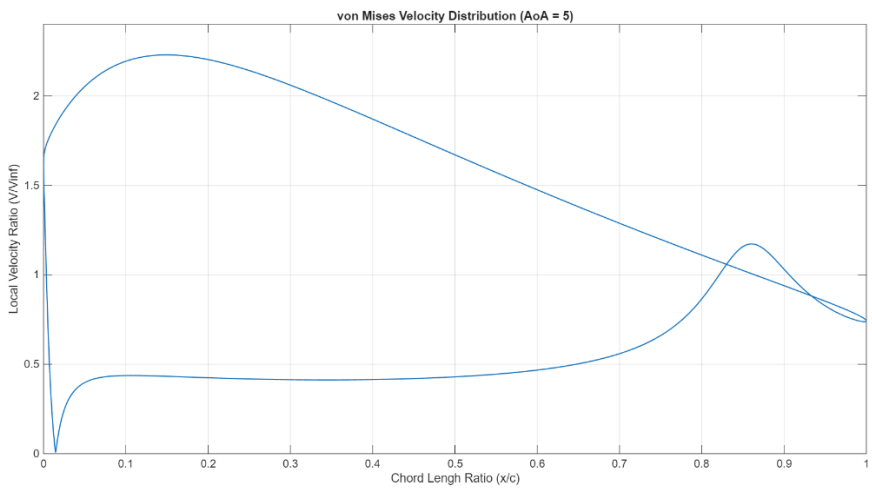


*Figure 10. Velocity Ratio at Angle of Attack of 5 Degrees.*

Figure 10 demonstrates the velocity ratio across the von Mises airfoil, using an angle of attack of 5 degrees. The velocity ratio is calculated from Equation 3 as well.

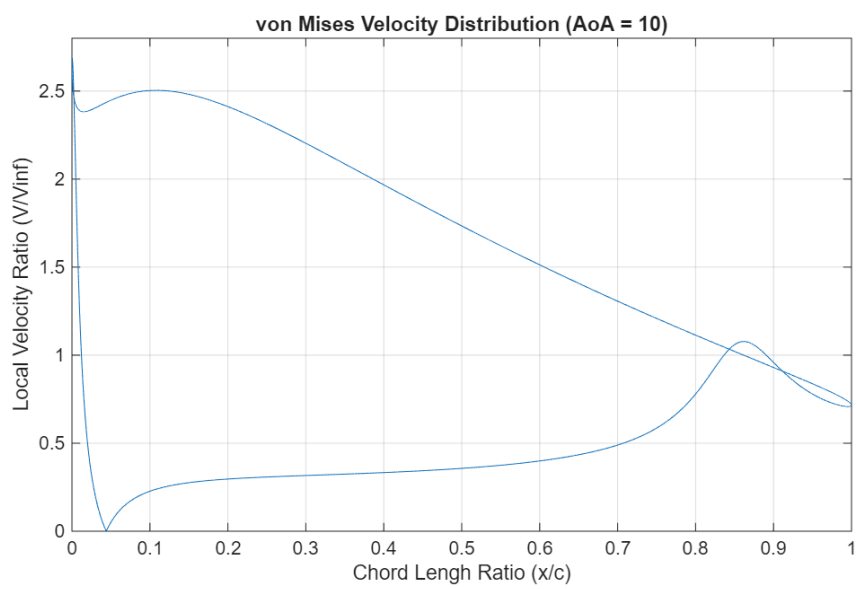


*Figure 11. Velocity Ratio at Angle of Attack of 10 Degrees.*

Figure 11 shows the velocity ratio across the von Mises airfoil at an angle of attack at 10 degrees, utilizing equation 3 once more.

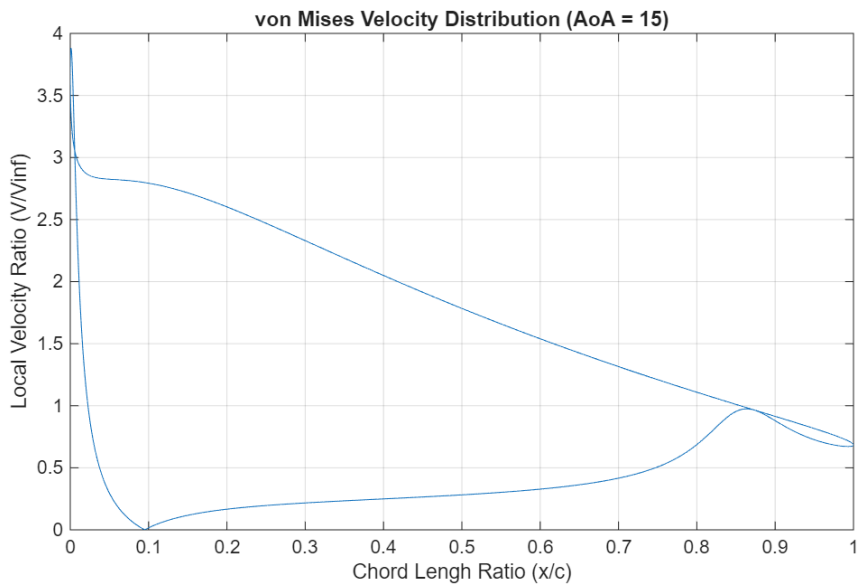


*Figure 12. Velocity Ratio at Angle of Attack of 15 Degrees.*

Figure 12 demonstrates the velocity ratio across the von Mises airfoil at an angle of 15 degrees, using Equation 3 as well.

## Appendix

```
% Calculate Circle Center and Radius from Thickness and Camber Ratio
camber_ratio = 0.05;                                % Camber ratio
beta_input = atan(2*camber_ratio);                  % beta value for Joukowski
transform
thickness_ratio = 0.15;                             % Thickness Ratio
epsilon = thickness_ratio/1.299;                    % epsilon for center circle
b = 0.25;                                           % b value for Joukowski transform
mu = (-epsilon*b) + 1i*(b*tan(beta_input));         % Center circle
a = abs(b - mu);                                    % Circle radius

% Geometry of Circle to be Transformed
theta_TE = angle(b - mu);                           % Theta at trailing edge
theta = linspace(theta_TE, theta_TE + 2*pi, 1000);
zeta = mu + a*exp(1i*theta);

% Plotting Circle
figure
plot(real(zeta),imag(zeta), 'b-')
hold on
plot(real(mu), imag(mu), 'b*', 'MarkerSize', 10, 'LineWidth', 2);
joukowski_zeros = [b, -b];

plot(real(joukowski_zeros), imag(joukowski_zeros), 'bo', 'MarkerSize', 8, 'LineWidth',
2);
hold off;
axis equal
grid on;
xlabel('x')
ylabel('iy')
title('Geometry of Circle to be Transformed (Joukowski)')

% Joukowski Airfoil Transformation
z = zeta + (b^2 ./ zeta);                           % Airfoil in Z plane
xLE = min(real(z));                                 % Leading edge in real plane
xTE = max(real(z));                                 % Trailing edge in real plane
c = xTE - xLE;                                      % Chord length
xc = (real(z) - xLE) / c;                           % x/c
yc = imag(z) / c;                                   % y/c

figure                                              % Figure 2
plot(xc,yc)                                         % Plot airfoil
xlim([0 1])
axis equal
grid on;
xlabel("Scaled Chord Length Position (x/c)")
ylabel('Scaled Thickness (y/c)')
title('Scaled Joukowski Airfoil')

% Velocity Destribution
beta_geom = -theta_TE;                              % beta for trailing edge

% Angle of Attack = 0
alpha = (pi/180)*0;

% Calculate v/vinf
V_s_Vinf = (2*abs ( (sin(alpha+beta_geom)) - (sin(alpha-theta)) ) );
dz_dzeta = (abs(1-(b^2 ./ zeta.^2)));
V_Vinf = V_s_Vinf ./ dz_dzeta;

% Fix the trailing edge (TE) points
```

```
V_TE = (b/a) * abs(cos(alpha + beta_geom));          % kutta condition
V_Vinf(1) = V_TE;                                    % first point
V_Vinf(end) = V_TE;                                  % last point

% Plot
figure
plot(xc,V_Vinf)
xlim([0 1])
ylim([0 1.6])
xlabel('Chord Lengh Ratio (x/c)')
ylabel('Local Velocity Ratio (V/Vinf)')
title('Nondimensional Velocity Ratio Across Airfoil (AoA = 0)')
grid on;

% Von Mises Airfoil Transformation

% R = 1.111, UR=-0.072, UI=0.290 from von mises check case
a = 1.111;
mu = -0.072 + 1i*0.290;
zT = 1.0;

% Geometry of Circle to be Transformed
theta_TE = angle(zT - mu);
theta = linspace(theta_TE, theta_TE + 2*pi, 1000);
zeta = mu + a*exp(1i*theta);

% Kutta condition
beta_geom = -theta_TE;

% Von Mises Circle
figure
plot(real(zeta), imag(zeta), 'b-')
hold on;
plot(real(mu), imag(mu), 'b*', 'MarkerSize', 10, 'LineWidth', 2);

z2 = -1.0 + 0i;
z3 = 0.370 - 0.471i;
z4 = -0.370 + 0.471i;
vm_zeros = [zT, z2, z3, z4];

plot(real(vm_zeros), imag(vm_zeros), 'bo', 'MarkerSize', 8, 'LineWidth', 2);
hold off;
axis equal
grid on;
xlabel('x')
ylabel('iy')
title('Geometry of Circle (von Mises)')

% zeta values from Von Mises check case
z2 = -1.0 + 0i;
z3 = 0.370 - 0.471i;
z4 = -0.370 + 0.471i;

% Calculate c1,c2,c3 coefficients
c1 = zT^2 - z2*(z3+z4) - z3*z4;
c2 = 0.5 * (zT*(z2*(z3+z4) + z3*z4) + z2*z3*z4);
c3 = (-1/3) * (zT*z2*z3*z4);

% Von Mises Transformation
z = zeta + c1./zeta + c2./(zeta.^2) + c3./(zeta.^3);

% x/c and y/c normalization
xLE = min(real(z));
```

```
xTE = max(real(z));
c = xTE - xLE;
xc = (real(z) - xLE) / c;
yc = imag(z) / c;

% Plotting
figure
plot(xc,yc)
axis equal
grid on;
xlabel("Scaled Chord Length Position (x/c)")
ylabel('Scaled Thickness (y/c)')
title('Scaled von Mises Airfoil')

% Velocity Distribution

% Angle of Attack = 5
alpha = (pi/180)*5;

% Numerator for V/Vinf
V_s_Vinf = (2*abs ( (sin(alpha+beta_geom)) - (sin(alpha-theta)) ) );

% New denominator for V/Vinf
dz_dzeta_factored = (1 - zT./zeta) .* (1 - z2./zeta) .* (1 - z3./zeta) .* (1 -
z4./zeta);
dz_dzeta = abs(dz_dzeta_factored);

% V/Vinf calculation
V_Vinf = V_s_Vinf ./ dz_dzeta;

% Numerator for Kutta condition
V_TE_numerator = 2 * abs(cos(alpha + beta_geom)) / a;

% Denominator for Kutta condition
zT_pp = (1/zT) * (1 - z2/zT) * (1 - z3/zT) * (1 - z4/zT);
V_TE_denominator = abs(zT_pp);

% Kutta condition
V_TE = V_TE_numerator / V_TE_denominator;

% First and Last Elements
V_Vinf(1) = V_TE;
V_Vinf(end) = V_TE;

% Plotting
figure
plot(xc,V_Vinf)
xlim([0 1])
ylim([0 2.4]) % Kept your specified ylim
xlabel('Chord Lengh Ratio (x/c)')
ylabel('Local Velocity Ratio (V/Vinf)')
title('von Mises Velocity Distribution (AoA = 5)')
grid on;
```